

Naga Pujitha Pitta

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SUMMARY

Machine Learning-oriented Computer Science graduate with research experience in model evaluation, statistical testing, and applied AI. Proven track record in developing and validating supervised learning models, anomaly detection systems, and explainable AI solutions across diverse domains. Strong foundation in experimental design, cross-validation, and comparative analysis of ML algorithms with emphasis on reproducibility and statistical significance.

EDUCATION

MS	Blekinge Institute of Technology Masters qualification plan in Computer Science 📍 Karlskrona, Sweden	Sept 2024 – June 2026(Expected)
BS	Blekinge Institute of Technology Bachelor qualification plan in Computer Science 📍 Karlskrona, Sweden	Aug 2023 – June 2024

INTERNSHIP

All India Council for Technical Education (AICTE) An internship was completed in AICTE In Python Programming. 📍 Kakinada, India	May 2023 – June 2023
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PUBLICATIONS

Analyzing Rule-Based Models in Personal Loan Credit Scoring Performed a comparative analysis of RIPPER and CN2 algorithms for personal loan credit scoring, assessing accuracy, precision, recall, and F1 score to evaluate their effectiveness in financial risk assessment. Tools used: Machine Learning	Jan 2024 - June 2024
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ACADEMIC PROJECT EXPERIENCE

Thesis - Technical Debt in Requirements Engineering Thesis - Technical Debt in Requirements Engineering

- Explored technical debt in requirements engineering, analyzing its sources during the requirements phase and strategies for effective management and mitigation. The research highlights the impact of early-stage decisions on project health and offers recommendations for addressing technical debt to enhance maintainability.

Deployment of multi-tier web application in AWS cloud.

- Deployed a multi-tier web application in the AWS cloud, utilizing services such as EC2, RDS, and S3 for scalable and efficient infrastructure management. Ensured secure and seamless integration of application layers to optimize performance and availability.
- Tools Used: AWS

Threat Modelling and Risk Analysis for LLM-based architecture.

- Conducted threat modeling and risk analysis for LLM-based architecture, identifying potential vulnerabilities and assessing risks to enhance system security. Developed mitigation strategies to address threats specific to large language model deployments, ensuring robust and secure operations.

Mobile App Development

- Text2Image, an application that converts text descriptions into high-quality images using AI. Implemented a user-

friendly interface and optimized algorithms for fast image generation, enhancing user experience through continuous improvements based on feedback.

- Tools used: flutter

Software Maintainability and Quality Analysis Using QM Framework

- Applied the Goal-Question-Metric framework to evaluate maintainability of large-scale software systems. Analyzed object-oriented metrics (LCOM, CBO, RFC, WMC, LOC, TCC, DIT) to identify technical debt and prioritize maintenance efforts for hard-to-maintain modules.

EV Charging Optimization Using Intelligent Predictive Decision Support Systems

- Designed predictive decision support models to optimize multi-horizon electric vehicle charging in smart cities. Implemented intelligent algorithms to balance energy demand, charging efficiency, and sustainability objectives.

Intelligent Decision Support System for Cryptocurrency Price Prediction

- Developed a machine learning-based system to predict short-term cryptocurrency price movements using high-frequency market data. Implemented and compared Random Forest, XGBoost, Gradient Boosting, SVM, and LSTM models with time-series preprocessing and feature engineering. Evaluated model performance using RMSE and MAE metrics.

Indian Stock Market Trend Analysis Using CRISP-DM Methodology

- Executed an end-to-end machine learning pipeline following CRISP-DM framework to forecast stock market trends. Performed exploratory data analysis, feature selection, and model comparison to support data-driven financial decision-making.

Accident Severity Prediction Using Weather and Road Infrastructure Data

- Built predictive models to estimate accident severity based on weather conditions and road infrastructure features. Conducted comprehensive data preprocessing, feature engineering, and supervised learning model evaluation for safety analytics.

Wine Quality Classification Using Machine Learning

- Developed supervised learning models to classify wine quality using physicochemical attributes. Evaluated multiple algorithms using accuracy, precision, recall, and confusion matrix analysis.

Spam Detection Using Supervised Learning Algorithms

- Implemented text classification system to detect spam messages. Compared algorithm performance using precision, recall, and F1-score metrics for robust spam filtering.

SKILLS

Languages: C++, C, Java, SQL, Python, Machine Learning, Basics of AI and deep learning, R Programming, Natural Language Processing,

Data Analysis: Pandas, Numpy, Matplotlib, Seaborn

Machine Learning: Scikit-learn, Keras, TensorFlow

Other: Communication Skills, Team Management

AVAILABILITY

Seeking internships, thesis opportunities, and junior-level positions in computer science and software engineering within Sweden.

INTERESTS

Technical writing and research documentation | Analytical problem-solving | Exploring UI design and technology blogs | Mindfulness and creative learning